

IKT215 Assignment 3:

Function Approximation Classification, Error Estimation and Model Selection

*All assignments are mandatory. Assignments equally contribute to 50% of your final grade. **Please make sure to read the additional instructions provided towards the end.***

Dataset

The mfeat-factors dataset consists of features extracted from handwritten digits (0-9) from a collection of Dutch utility maps. The dataset is provided on the assignment page for Canvas as `*.npz`` files. Further information about the datasets is also available on Canvas.

Assignment Tasks

Part 1: Setup and Exploration

1. Load the mfeat-factors dataset.
2. Report the number of samples, number of features, and number of classes.
3. Prepare the data (normalization).

Part 2: Neural Network Implementation

1. Implement a fully connected neural network (You may use any framework of your choice).
2. Document your network architecture and training approach.

Part 3: Error Estimation

Implement and compare the following error estimation techniques:

- Resubstitution error estimation
- Hold-out set error estimation
- 5-fold cross-validation

For each method:

- Report the error estimates
- Briefly discuss the advantages and limitations.

Part 4: Model Selection

Implement and compare the following approaches for model selection:

- Train-validation-test approach.
- Cross-validation approach

For each approach:

- Test at least two network configurations.

- Document how you selected the final model.
- Report final performance on test data.

Additional Instructions

1. **Prepare a report (6 pages maximum) containing:**
 - Introduction
 - Implementation details,
 - Results comparison,
 - Key insights about the methods.
2. You don't need to submit your code.
3. The report should include everything mentioned in the task description.
4. Make sure to report any hyperparameter choices you made.
5. Only include the relevant visualizations in the report.
6. The visualizations in the report should be legible.

Rubric

Task	Score (%)
Part 1	10
Part 2	20
Part 3	30
Part 4	30
Report quality and structure	10
Total	100

Use of Generative AI:

While the use of generative AI tools is not prohibited, it is strongly discouraged for this assignment. All reports must include a **dedicated disclaimer page** that honestly describes which AI tools (if any) were used and how they were applied, in accordance with [UiA's general guidelines](#). Remember that **submitting AI-generated text without proper citation constitutes plagiarism and will be treated as such.**

Late Submission Policy

Late submissions will incur a **5% penalty per calendar day**. Submissions will not be accepted more than six days after the deadline. Plan accordingly and contact the instructor in advance if you anticipate difficulties meeting the deadline.